JAYDEN METCALFE

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TECHNICAL SKILLS:

Languages: Python, TypeScript, JavaScript, Java, C, Bash, Clojure
Frontend: React, Angular, HTML, CSS, TailwindCSS, React Testing Library, Vitest, Next.js
Backend: Node.js, Express.js, FastAPI, OAuth 2.0, Jest
Databases: PostgreSQL
Cloud & DevOps: AWS, Vercel, Docker, REST APIs
Machine Learning & Data: scikit-learn, Keras/TensorFlow, Pandas, NumPy, NLTK, Matplotlib, lingpy
Tools: Git, GitHub, Arduino, Linux, Figma, Claude

WORK EXPERIENCE:

Full Stack Software Developer & U16 Alpine Ski Racing Coach November 2024 - present
Whistler Mountain Ski Club, Whistler BC

- Engineered a full-stack analytics web application using React, TypeScript, Node.js, Express.js, PostgreSQL, and Python to track and analyze athletes' training data, implementing secure OAuth 2.0 authentication and role-based access control
- Architected and optimized RESTful APIs and SQL queries to support filtering and comparative analytics of hundreds of data points, improving query performance and enabling data-driven adjustments to team training programs
- Containerized using Docker and Docker Compose to separate frontend, backend, and database services, improving scalability and maintainability, then deployed the application as an Amazon Web Services (AWS) EC2 instance
- Developed unit and integration test suites using Jest, Vitest, and React Testing Library, improving application reliability
- Designed a Python program to extract, filter, and visualize timing data, generating dynamic charts and reports to analyze athlete trends, progression, and performance comparisons across demographic metrics including age and gender
- Building a custom Uniform Manager platform with Next.js, TypeScript, PostgreSQL (Neon), and Prisma to streamline the distribution and tracking of hundreds of equipment pieces across 45+ employees, leveraging Claude AI as a development assistant to accelerate feature delivery

Reinforcement Learning Researcher August 2023 - May 2024
The Britt Lab, Montréal QC

- Automated data processing and reporting using Python and Excel, generating visualizations from thousands of data points to support behavioural research analysis
- Implemented Q-learning models to simulate and analyze decision-making behaviour
- Debugged and extended Arduino code to scale for multiple experimental stages, each with variables that required fine tuning to maximize the collection of usable data points

Product Research, Design, and Development Intern May - August 2023
Interac Corp., Toronto ON

- Prototyped RESTful API endpoints and a PostgreSQL database schema for an escrow-based e-transfer solution to enhance transaction security and reduce fraud risk, which was demonstrated to senior leadership
- Collaborated with cross-functional teams to build and update frontend features of an emulator using TypeScript and Angular within an MVC architecture, implementing UI/UX designs from Figma for client-specific product demos
- Refactored legacy code to improve maintainability, scalability, and integration with new product features

PERSONAL PROJECTS:

FIS Race Points Calculator | *Python, Node.js, JavaScript, HTML/CSS, REST APIs*

- Developed a full-stack web application using Python, Node.js, and REST APIs to calculate ski race scores in real time, reducing processing time from several hours to seconds
- Integrated external APIs to process live timing data and automate scoring calculations
- Built a responsive frontend using HTML, CSS, and JavaScript to display real-time results

EDUCATION:

Bachelor of Arts and Science, Cognitive Science August 2020 - April 2024
McGill University, Montréal QC, Canada

- Neuroscience stream, Minor in Computer Science

Relevant Coursework: Algorithms & Data Structures, Python, Java & OOP, Probability, Software Systems, Computational Linguistics, Discrete Structures, Applied Machine Learning, Natural Language to Data Science, Linear Algebra, Deductive Logic